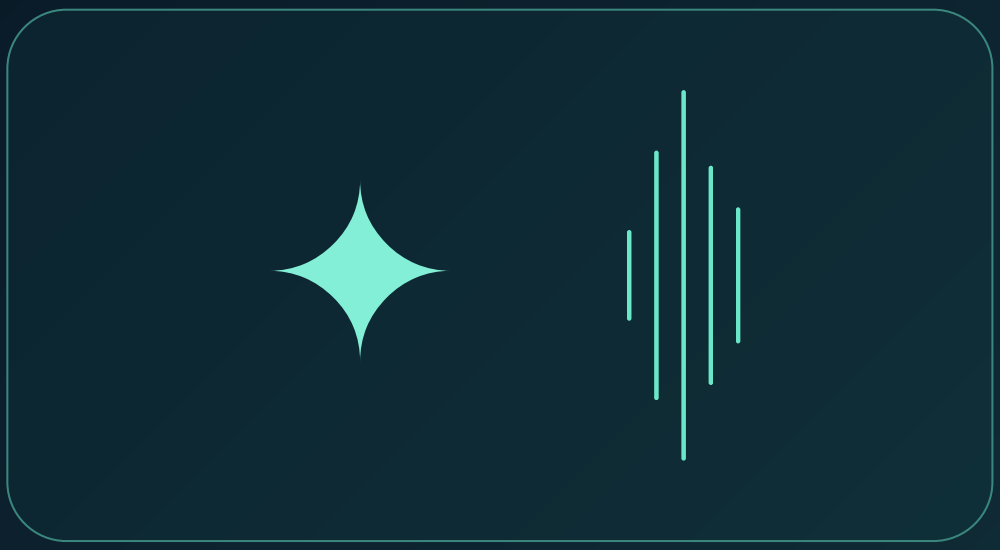
MEETSAATHI / PRODUCT EXPERIENCE SYSTEM



CONVERSATION COMPANION

The complete product experience

Present during every conversation.
Intentional about every memory.

Product & Engineering Specification

18 October 2026 · Design baseline v1.0

Not a functioning native application

Version: 1.0 design baseline | **Date:** 18 October 2026 | **Status:** Product and implementation handoff; not a built mobile app

> ***Product promise:*** *Be more present in everyday conversations. Understand what matters. Remember only what you choose to keep.*

1. Executive product decision

Conversation Companion is the general-purpose, consent-aware mode within **one unified MeetSaathi mobile application**. It is neither a secret recorder nor a social-manipulation engine. It helps people prepare, listen, clarify, remember, and reflect across permitted everyday conversations. It shares the app's audio, reasoning, voice, memory, and privacy foundation with Meeting, Learning, and Interview Companions; only the experience policies change.

Design thesis: The best everyday companion is calm, relationally respectful, personally useful, and often silent. It should feel like a helpful thinking partner, not a second person competing for attention.

Key difference from Meeting Companion: A meeting often produces decisions and action items. An everyday conversation may produce an understanding, a personal commitment, emotional nuance, a new idea, or nothing worth saving. Conversation Companion therefore asks *what is worth keeping?* instead of automatically turning every exchange into a business-style recap.

Success definition: A user can initiate a permitted session in under three deliberate steps, keep their attention on the other person, retrieve one precise and well-attributed useful fact, and decide exactly what—if anything—is retained afterward.

2. Users, environments, and fit

Situation	Real job to be done	Desired AI intervention	Risk and constraint
Mentor or career conversation	Remember advice, clarify next steps	Highlight an unanswered question; save a follow-up	Respect third-party confidentiality
Networking or industry event	Remember permitted introductions and topics	Quiet context recall when requested	Do not covertly profile strangers
Collaborative problem-solving	Connect ideas and capture a useful breakthrough	Surface a trade-off or a contradiction with evidence	Don't claim agreement prematurely
Trusted family discussion	Remember mutually agreed plans	Optional save on user request	Highest privacy default; no background surveillance
Language exchange	Understand difficult phrases	On-demand translations and context	Preserve meaning and indicate uncertainty
Personal coaching session	Recall guidance and exercises	Save user-selected tasks	Sensitive topics; avoid unverified therapy/diagnosis
Medical / legal / financial consultation	Remember practitioner-provided instructions	Evidence-linked user questions	Stronger domain controls; no substituting for qualified advice

Situation	Real job to be done	Desired AI intervention	Risk and constraint
After-the-fact private reflection	Think through a conversation from memory	Organize user's notes and brainstorm	Must label recollection as user-supplied, not a transcript

Audience priority: Everyday professional, mentoring, and collaborative conversations first; high-stakes personal and regulated situations later, only with domain-specific safeguards and deployment approval.

Not supported: Secretly recording people; detecting someone’s inner feelings or intentions from their voice; creating covert dossiers about others; impersonating an individual; making consequential judgments about people; manipulating another participant. A permitted coaching prompt must help the user communicate clearly, not engineer deceptive persuasion.

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3. North-star journey

1. **Choose to start.** The person launches MeetSaathi and selects *Conversation*, which defaults to **Notes-only** unless the user deliberately enables microphone capture and meets participant-notice and consent requirements.
2. **Preflight.** The app indicates the intended capture source, voice destination, network status, retention setting, and a concise description of the participant-awareness obligation.
3. **Begin.** The session indicator unambiguously shows `Listening` or `Notes-only`. The user can pause or stop without navigating away.
4. **Face-up live companion.** The dashboard displays the current theme and at most one high-value, ephemeral suggestion. *Ask*, *Save moment*, *Pause*, and *End* remain available.
5. **Pocket companion.** The person locks the screen, if the platform permits an active session to continue. Optional earbud guidance remains short and never falls back to loudspeaker after headset disconnection.
6. **Follow a thread.** When the user deliberately asks or when a validated intervention qualifies, the AI gives a question, clarification, or evidence-backed reminder—not a long unsolicited lecture.
7. **Review and choose.** The user ends the session and sees a private review of potential keeps; nothing becomes durable cross-session memory without the selected retention policy and authorization.
8. **Reflect.** The user can ask `What was the suggested next step?` or `Help me think through this idea`. The system separates **said in recording**, **my note**, and **AI interpretation**.
9. **Prepare next time.** Saved information resurfaces only inside the user's allowed memory spaces and with context.

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4. Information architecture

Unified app bottom navigation: **Home · Live · Memories · You**. *Conversation* is an experience preset inside Home/Live, not a new app or competing tab. In a live session, bottom navigation is replaced by high-priority session controls to avoid accidental exits.

Conversation screens and requirements:

ID	Screen	Primary information	Primary action	State / behavior
C01	Companion Home	Continue / start, recent saved topics	Start conversation	No mic on launch
C02	Context chooser	Mentor, personal, networking, open	Choose context	Defaults are preferences, not permissions
C03	Capture & consent	Recording options, notice, retention	Confirm session	Microphone gated
C04	Audio preflight	Input, output, signal, route warning	Begin session	Block on invalid route or missing permission
C05	Live Focus	One insight, current topic, statuses	Ask / Save / Pause	No auto-scroll or endless cards
C06	Live transcript	Provisional speaker segments and time	Inspect / correct	Optional and collapsible
C07	Ask Companion	Current context and question	Ask	Answer may be silent text or private voice
C08	Earbud / pocket	Audio route, privacy voice, interruptions	Adjust or mute	Never loudspeaker fallback for private speech
C09	Saved moments	Bookmarks / notes with provenance	Bookmark / remove	No false certainty
C10	Session pause/end	Capture status and retained draft	Resume / End	Idempotent end
C11	Reflection recap	Themes, keeps, questions, source clips	Review and save	Private by default
C12	Reflection chat	Remember / Think / Prepare	Ask grounded questions	Modes separate evidence from reasoning
C13	Conversation memory	Scoped topics, dates, saved facts	Search	Space and permission aware
C14	People & boundaries	Participant aliases, sharing status	Correct labels	No covert profiles or identity guesses
C15	Next-time prep	Voluntary relevant memory summary	Open prep	Never surface sensitive info on lock screen
C16	Privacy center	Delete, export, retention, audience	Manage	Per-session memory control
C17	Notes-only mode	User-authored notes, no capture	Capture note	No mic icon or fictional transcript
C18	Recovery & limited mode	Interruption / offline / route changes	Retry / save locally / end	Truthful degraded statuses

5. Design language and UX requirements

- **Visual direction:** warm off-white surfaces for reviewing knowledge; deep midnight navy for active conversations; calm jade as a signal of live/ready state; amber only for unresolved or caution states. Respect system appearance, large type, reduced motion, and high contrast.
- **Home:** strong hierarchy: `Start Companion` > `Continue Reflection` > `Memories`. Do not turn the home screen into an analytics dashboard.
- **Live screen:** one active insight max. No animated paragraph streaming. Transcript folded by default; expand only on demand.
- **Tap targets:** minimum 44×44pt iOS / 48dp Android for critical actions. End requires a visible control, not gesture discovery.
- **Information architecture:** label user-authored, captured, inferred, and independently researched content consistently.
- **Live suggestion length:** aim for 12–28 words; hard design cap 45 words before `More`. Never issue multi-paragraph private audio guidance during ongoing speech.
- **User control:** separate `Capture`, `Proactive suggestions`, `Voice playback`, and `Memory retention`. Turning off one must not deceptively imply that all are off.
- **Accessibility:** captions, Dynamic Type, TalkBack/VoiceOver, color-independent status, motion settings, RTL where appropriate, and no audio-only essential action.
- **Sound:** no auto-read aloud, no public-speaker playback by accident, earbud voice intentionally enabled only.

6. Live Focus: component contract

Header: `Conversation · Live`, elapsed timer, exact session status, active input/output icons. **Current topic:** short provisional heading updated cautiously; no labeling a participant's emotional state. **One insight:** type (`Clarify`, `Connection`, `Remember`, `Question`), body, `Why this?` rationale, `Save`, `Dismiss`. **Quick actions:** `Ask`, `Save moment`, `Pause`, `End`. **Transcript drawer:** collapsed, source audio markers when authorized. **Accessibility:** persistent visible and spoken recording status. **Privacy microcopy:** `Participants have been informed` is a user-confirmed state, not an assertion made by the AI.

When there is no useful insight, show `Listening quietly — ask whenever you need` rather than forced content. When audio quality falls, show `I may have missed part of this conversation` and suppress speaker-specific claims. When notes-only, replace the entire listening dashboard with a note composer; do not claim live awareness.

7. Proactive intelligence and intervention policy

Pipeline: consent gate → capture / ASR (when permitted) → provisional speaker turns → current conversational context → candidate generation → grounding / safety / usefulness checks → interruption scheduler → text/voice decision → user feedback → memory eligibility.

Allowed intents: (a) clarify an ambiguity, (b) recall user-authorized relevant context, (c) propose a respectful open-ended question, (d) flag a missed explicit follow-up, (e) explain an unfamiliar term on request, (f) translate with caveats, (g) highlight a concrete inconsistency backed by source evidence.

Disallowed intents: guess emotions or hidden motives; provide manipulative interpersonal tactics; make definitive claims about trustworthiness or character; leak another person's private prior

conversation; present AI advice as quoted speech; infer protected characteristics; propose unrelated suggestions purely to boost engagement.

Intervention rules (initial design hypotheses):

1. `Silent` default for sensitive contexts, `Balanced` for explicitly opted-in ordinary conversations, `Active` only when selected. No preset overrides consent.
2. Minimum independent-evidence / context-quality checks; if low-confidence, abstain or offer an optional question to clarify.
3. Suppress delivery while the user is speaking; consider pause timing, meeting pace, and whether the new message is better saved for recap.
4. Deduplicate semantically similar suggestions. Default cooldown 5 minutes in Balanced, unless a genuinely critical user-defined reminder is due. Not a fixed limit on all emergency or accessibility behaviors.
5. Text is the default presentation on face-up devices; voice requires explicit authorization and a verified private route.
6. Competing candidate insights ranked by `usefulness × evidence × timeliness − distraction × sensitivity`; qualitative score with separately testable evidence and privacy gates, not unexplainable single-model threshold.
7. Every delivered insight records `kind`, `source refs`, `model/policy version`, `delivered_at`, `user reaction`, and `reason to intervene` without storing more participant data than necessary.
8. Do not continuously surface intervention cards just because they can be generated. The app earns trust through restraint.

Examples:

- Mentoring: `You said you want clearer priorities. Consider asking which skill to focus on first.`
- Language exchange: `The phrase may mean “let's revisit it later,” depending on context. Tap to see alternatives.`
- Collaborative idea: `Two timelines were mentioned. Ask which one is the actual commitment.`
- Personal conversation: `You asked to remember the appointment date. Save the date?` (Only if source clear and within capture permissions.)
- No suggestion: `Keep listening`; the most common outcome should often be no interruption.

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8. Ask Companion: reasoning contracts

Three tabs with separate retrieval and response rules:

- **Remember:** `What was said?` Answer only from authorized transcript or the user's notes, with `Source: recording at 08:24` / `Source: my note, unverified` and `Not enough evidence` when applicable.
- **Think:** `What could it mean?` Analyze alternatives, uncertainties, respectful questions, and user goals; explicitly label AI inference. Never assert another person's hidden motive as fact.
- **Prepare:** `What can I do next?` Suggest follow-up questions, optionally a user-reviewed reminder or private to-do. No autonomous messaging or calendar actions without distinct permission.

When no recording exists: allow questions about typed or privately dictated notes. Do not invent direct quotes or timestamps. When a transcript conflicts with a user correction, preserve the original, the correction, and the confidence/provenance relationship.

9. Private voice / pocket experience

- User starts from visible app, routes audio deliberately, and confirms participant awareness when live capture is enabled.
- Android: microphone foreground service with proper permission / service type; the session must be initiated under allowed foreground conditions. Persistent system capture indicator/notification.
- iOS: correct `AVAudioSession` category and background audio mode; handle interruptions and route changes. A compatible device may still not simultaneously support the desired phone microphone and Bluetooth voice output; detect and explain actual routing.
- Bluetooth HFP routing may couple the headset microphone and output on iOS, meaning the earbud microphone may replace the table microphone. Never promise remote participants will remain audible when the mic changes.
- Voice uses brief interjections, optional earbud chime/haptic, voice interruptibility, and queuing. If headset disconnects, **mute private spoken content immediately**, show an alert, and do not reroute private speech to speakerphone.
- If a session's input fails or the operating system suspends it, surface `Capture interrupted`; save only authenticated captured chunks and do not silently infer ongoing speech.
- Pocket audio capture is environment dependent, especially fabric noise, distance, and orientation. Offer `Audio quality poor` rather than suggesting a falsely perfect transcript.

10. Memory: store meaning without building a dossier

Memory is **opt-in, scoped, editable, portable, and forgettable**. Spaces: `Personal`, `Work`, `Learning`, and optional user-created spaces. The same speaker alias across sessions is *not* automatically the same person. Identity mappings require a transparent user-mediated workflow and relevant participant permission/legitimate basis where applicable.

Memory record kinds: `explicitly saved fact`, `question`, `idea`, `commitment`, `user note`, `possible topic`, `correction`, and `AI reflection`. Every item retains `source_type` = recorded / user_note / model_analysis; `session_id`, time interval when recorded, sensitivity, audience, expiration, provenance, and revision chain. AI analysis is not searchable as a quote from a participant.

Review-to-save: at end, offer `Keep these 2 things`, `Review individually`, `Discard session`. Default retention must be visible. Sensitive contexts default to ephemeral/notes-only when possible. Users can delete audio while retaining user-reviewed sanitized notes; clarify implications of doing so, including the loss of source playback.

Cross-conversation retrieval: permit only from a space the user has authorized for the current query. No surprise references from personal memory in a shared/work meeting. Search returns provenance, date, and original wording when available. Deletion cascades to embeddings, indexes, summaries, cache, and derived items or marks them inaccessible pending bounded deletion completion.

11. Session state machine

`IDLE → CONFIGURING → PREFLIGHT → ACTIVE_CAPTURE | ACTIVE_NOTES_ONLY → PAUSED → (ACTIVE_*) → ENDING → FINALIZING → REVIEW → SAVED | DISCARDED`

Exception sub-states: `INPUT_UNAVAILABLE`, `ROUTE_CHANGED`, `NETWORK_DEGRADED`, `BACKGROUND_RESTRICTED`, `CONSENT_NOT_CONFIRMED`, `STORAGE_UNAVAILABLE`, `PARTICIPANT_WITHDRAWS`, `PROCESSING_FAILED`.

Invariants:

1. No microphone access or streaming outside confirmed authorized capture states.
2. Notes-only never starts ASR or generates live listener-derived insights.
3. All `End` operations idempotent and release active capture resources immediately.
4. Session stop halts new capture before any background finalization begins.
5. Loss of consent stops recording for the affected context; do not resume automatically.
6. Voice output requires an approved destination; route change invalidates approval until re-evaluated.
7. Failure notices must use plain language and preserve the user's ability to end.
8. Saved and discarded statuses must accurately reflect committed storage, not optimistic UI state.

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12. UI state and event contracts

Representative event stream (`conversation_event_v1`):

```
{
  "event_id": "evt_123",
  "session_id": "sess_123",
  "sequence": 218,
  "kind": "insight_candidate|insight_delivered|utterance_final|audio_route_changed|capture_interrupted",
  "source_refs": [{"utterance_id":"utt_52","start_ms":120500,"end_ms":128100}],
  "privacy_space_id": "space_personal",
  "created_at": "2026-10-18T15:00:00Z",
  "payload": {"confidence_band":"moderate","text":"Clarify which date is confirmed."}
}
```

Represent consent authorization separately from OS microphone permission:

```
{
  "audio_capture": "permitted|not_permitted|pending",
  "participant_notice": "confirmed|not_confirmed|unknown",
  "ai_live_assistance": "enabled|disabled",
  "voice_output": "disabled|private_route_confirmed|route_unavailable",
  "retention": "ephemeral|session_review|user_selected_memory",
  "current_input": "phone_mic|headset_mic|external_mic|none",
  "current_output": "screen_text|private_headset|muted"
}
```

Client contract examples:

- `POST /v1/conversations` — create private draft, no recording yet.
- `POST /v1/conversations/{id}/authorize` — verify client disclosures and permissions; server does not assume third-party legal consent from a checkbox alone.
- `POST /v1/conversations/{id}/start` — idempotency key, route / privacy preflight.
- `POST /v1/conversations/{id}/pause` / `/resume` / `/end` — monotonic state transitions.
- `GET /v1/conversations/{id}/events?after_seq=...` — resumable context updates.
- `POST /v1/conversations/{id}/ask` — mode = remember/think/prepare; explicit retrieval scope.

- `POST /v1/conversations/{id}/bookmarks` — note / clip / correction with provenance.
- `POST /v1/conversations/{id}/review` — atomic save selection and retention.
- `DELETE /v1/conversations/{id}` — deletion workflow with status and downstream propagation.
- `GET /v1/memory/search` — scoped, permission-filtered evidence.

12.1 Core entities

`Account`, `PrivacySpace`, `Session`, `SessionPermissionSnapshot`, `ParticipantNotice`, `AudioChunk`, `Utterance`, `SpeakerAlias`, `Insight`, `InterventionDecision`, `Bookmark`, `MemoryItem`, `MemorySource`, `MemoryCorrection`, `RetentionPolicy`, `DeletionRequest`, `AuditEvent`.

Use stable IDs, tenant/user ownership on every row, encryption metadata, versioned transcript revisions, and explicit speaker uncertainty. Index consent and privacy scope checks before vector retrieval, not merely after semantic ranking.

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13. Production implementation architecture

Native applications: Kotlin + Jetpack Compose on Android; Swift + SwiftUI on iOS. Shared OpenAPI and event schemas; separate platform audio modules. Why: live microphone, interruptions, Bluetooth, accessibility, and lifecycle control are platform-sensitive.

Client components: SessionController, PermissionCenter, AudioCaptureAdapter, AudioRouteDirector, LocalEncryptedBuffer, LiveEventReducer, CompanionDashboard, VoicePlaybackCoordinator, ReflectionUI, MemoryBrowser, RecoveryManager.

Backend components: API Gateway, Session Service, Stream Ingest, ASR & diarization workers, Turn Alignment, Context Builder, Evidence Store, Intervention Policy, LLM Router, TTS, Memory Service, Retrieval, Retention/Deletion, Audit and Observability.

Data: PostgreSQL for transactions, scoped relationship and correction data; pgvector initially for embedding search; encrypted object storage for permitted audio; Redis only if workload measurements justify it; queue for durable asynchronous finalization. Avoid premature microservice / graph / Kafka expansion.

Realtime processing: modest streaming chunks → VAD → streaming ASR → speaker turn alignment → evidence indexed → context engine → relevance/safety judge → delivery decision. Batch reprocessing after recording may improve final speaker labels and transcripts; never silently overwrite originals.

On-device-first ambition: privacy-sensitive local transforms and buffering first, selectively use on-device speech inference where latency/battery/quality supports it; cloud processing requires disclosure and a recorded permission decision. Do not imply Nemotron 3’s published buffer latency equals end-to-end assistant latency.

Failure recovery: resumable event sequences, bounded encrypted local buffer, explicit capture-incomplete status, signed upload grants, safe replay and deduplication. No fake 'always listening' promise after an OS stop.

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14. Trust and security threat model

Threats and mitigations:

Threat	Required control
Unauthorized recording	Foreground start, OS permissions, explicit capture opt-in, participant disclosures, visible indicators
Listening when paused	Mic stream teardown / verified no bytes sent
Voice leaking to phone speaker	Route latch and automatic private TTS mute on route change
Mixing personal and work memory	Privacy-space ACLs checked at retrieval and generation
AI claims wrong speaker identity	Anonymous ID defaults, correction UX, evidence with confidence
Prompt injection spoken by another person	Treat transcripts as untrusted, never as system instructions; tool execution requires policy + approval
Transcript containing private secrets	Redaction controls, retention controls, narrow access, encrypted storage
Cross-user search leakage	Tenant-scoped authorization before retrieval; negative tests
Deletion appears complete but embeddings remain	Deletion jobs with auditable completion and bounded SLA
Sensitive notifications on lock screen	Neutral notification content by default
Unapproved autonomous actions	Explicit user confirmation before sending, scheduling, or sharing

Jurisdiction: Consent and data-protection bases vary by region and environment. Global rollouts need localized legal review; do not assume one tap solves legal authorization. For minors or sensitive settings, add dedicated safeguards, institutional rules, and stricter retention.

15. Quality and experience gates (targets, not achieved metrics)

Metric	Candidate release gate	How measured
Task completion: start legitimate session	≥ 95% without help in usability testing	Moderated tasks
Silent mode surprise outputs	0	Device automation + field QA
Lost audio on supported 60-minute sessions	< 0.5% unannounced	Long-session test matrix
False confident speaker attribution	0 in high-risk evaluation sample	Human-audited clips
Source provenance on factual remembered claims	≥ 98% correct in evaluation corpus	Evidence annotation
Proactive insights judged helpful	≥ 80% in opted-in, non-sensitive study	Blind human rating
Unwanted Balanced interruptions	≤ 1 per 30 minutes	Session feedback + review
Private audio leaks after disconnect	0	Bluetooth route fault injections

Metric	Candidate release gate	How measured
User-selected deletion	Complete within published service policy	End-to-end deletion check
Crash-free sessions	≥ 99.8% supported device set	Telemetry
Latency to first live text after utterance	P95 ≤ 3 s target	Timestamped end-to-end test
Relevant suggestion from trigger	P95 ≤ 6 s target	Session replay tests

Targets must be revised only with recorded user research, not to conceal a miss. Include accuracy slices for quiet rooms, café noise, phone-in-pocket, headsets, overlap, accents, languages, and network conditions.

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16. Analytics and user experience evaluation

North Star: **Meaningful companion conversations per weekly active user**—a session is meaningful only if the person confirms help received or returns to valuable saved context, not simply because recording stayed on. Guardrails: permission drop-offs, microphone errors, unnecessary interruptions, battery drain, retention surprise, memory trust, and deletion success.

Respect privacy: telemetry should use event metadata and opt-in aggregated quality measurements; never default to sending raw personal conversation content into product analytics.

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17. Delivery roadmap / engineering sequence

R1 — Coherent native journey: one Start Companion, transparent consent/preflight, Notes-only option, reliable pause/end, local encrypted capture, graceful interruptions. *Exit:* real device begins and finishes a valid session with no silent capture state change.

R2 — Accurate conversational understanding: streaming ASR, provisional speaker turns, collapsible transcript, confirmed corrections, source-linked capture, good degraded behavior. *Exit:* user retrieves an exact segment with timestamp and uncertainty.

R3 — High-utility intelligence: contextual Ask, one-card live focus, evidence-aware proactive scheduler, feedback, separate Remember/Think/Prepare answers. *Exit:* robust human evaluation and low interruption rate.

R4 — Memory and reflection: review-to-save, scoped search, user edits, export and deletion, next-time preparation. *Exit:* correct cross-conversation search without leakage.

R5 — Private voice and expanded devices: verified Bluetooth routes, earbud playback, interruption/route recovery, screen-off and pocket performance. *Exit:* supported-device matrix meets audio leak and recovery requirements.

R6 — Product readiness: security review, accessibility, localization, country launch policies, store compliance, battery/thermal validation, performance budgets, 90-minute reliability suites. *Exit:* product gates passed and user testing demonstrates willingness to return.

R1–R6 are release increments of one product, not disposable prototypes. Engineering can work on components in parallel once interface contracts stabilize.

18. Copywriting examples and behavior tone

- Initial welcome: `A little help when it matters. Space to be present the rest of the time.`
- Permission explanation: `Record only when everyone who needs to know has been informed and the recording is allowed.`
- Quiet live: `Listening quietly. Tap Ask if you need me.`
- Suggestion: `A date was mentioned, but not confirmed. Want to clarify it?`
- Poor capture: `Audio is unclear. I may miss words or speakers.`
- Private route loss: `Earbuds disconnected. Spoken guidance has stopped.`
- No-recording: `Notes-only. I can't hear this conversation; add anything you'd like to remember.`
- After ending: `What would you like to keep from this conversation?`
- Unsourced answer: `I can't confirm that from the material you saved.`

19. Product decisions and rejected alternatives

Choice	Decision	Why	Alternative rejected
Privacy default	Notes-only until explicit recording choice	Everyday setting is sensitive	Auto-listening at app open
Default live UI	One insight and clear status	Reduces distraction	Infinite transcript/chat wall
Assistance	Optional Balanced / Silent; Active opt-in	Timely value beats quantity	Always-on unsolicited prompts
Voice	Private route required	Prevents unintended disclosures	TTS through whatever speaker is available
Attribution	Anonymous unless confirmed	Model cannot know person identity from diarization alone	Guess names from voice
Memory	Review / scoping / deletion	User agency and trust	Permanent hidden memory of everyone
Modes	Remember / Think / Prepare	Separates evidence from inference	One unlabelled chatbot answer
Architecture	Native app + shared backend	Correct platform control	All audio in webview
Quality	Targeted evaluation gates	Product excellence is measurable	Feature count as quality metric

20. Implementation checklist for a Codex or engineering team

- ☐ Establish monorepo with Android/iOS/backend/contracts/docs/QA; enforce type-safe contracts.
- ☐ Build unified design system tokens and components used by all MeetSaathi companions.
- ☐ Implement session and permission state machine with test-first invariants.

- ☐ Implement native mic capture and device-route reporting without any silent background auto-start.
- ☐ Add UI-visible truthful state, notifications, interruptions and persistent user stop access.
- ☐ Implement mock synthetic event stream for app tests, separated from production stream endpoints.
- ☐ Implement secure stream ingest, ASR, diarization/alignment, and stable evidence IDs.
- ☐ Implement Ask/Reasoning with grounded, source-aware user-permission-scoped retrieval.
- ☐ Implement candidate interventions, scoring, suppress/abstain path and feedback capture.
- ☐ Implement private voice coordinator with route-change mute and verified destination.
- ☐ Implement post-session review, memory opt-in, corrections, search, export and deletion.
- ☐ Implement feature flags and server-driven policy versions without reducing explicit consent requirements.
- ☐ Run reproducible real-device acceptance suites for power, Bluetooth, network and permissions.
- ☐ Complete privacy/security review, penetration test, accessibility audit, regional policy review.

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21. Source and verification notes

Official technical references consulted 18 October 2026:

- Android microphone foreground-service rules: https://developer.android.com/develop/background-work/services/fgs/service-types and https://developer.android.com/develop/background-work/services/fgs/restrictions-bg-start
- Apple Bluetooth Hands-Free audio routing: https://developer.apple.com/documentation/avfaudio/avaudiosession/categoryoptions-swift.struct/allowbluetooth
- Apple background audio: https://developer.apple.com/documentation/avfaudio/avaudiosession/category-swift.struct/record
- Apple App Review recording consent: https://developer.apple.com/app-store/review/guidelines/
- NVIDIA Nemotron 3 diarization: https://huggingface.co/nvidia/Nemotron-3-Diarization
- European Commission GDPR principles: https://commission.europa.eu/law/law-topic/data-protection/information-business-and-organisations/principles-gdpr_en

Research boundaries: This is a prospective design with measurable targets, not proof of real-device support, legal compliance, model quality, or product-market fit. Obtain local legal advice and user evidence before launching in a jurisdiction or claiming performance.

One MeetSaathi. Multiple contextual experiences. One promise: useful help without stealing the user's attention or autonomy.